

DAA Lab

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Topic: Kruskal's Algorithm

Kruskal's Algorithm

Code:

```
C:\Users\devan\OneDrive\Desktop> g++ DEMO.c -std=c++11 -O2
1 // Include headers
2 #include <iostream>
3 #include <vector>
4 // Structure for an edge
5 struct Edge {
6     int u, v, w;
7 };
8
9 // Disjoint Set
10 int parent[100];
11
12 // Find function (with path compression)
13 int find(int i) {
14     if (parent[i] == i)
15         return i;
16     return parent[i] = find(parent[i]);
17 }
18
19 // Compare function to sort edges by weight
20 int compare(const void *a, const void *b) {
21     return ((struct Edge*)a)->w - ((struct Edge*)b)->w;
22 }
23
24 void kruskal(struct Edge edges[], int V, int E) {
25     // 1. Sort edges by weight
26     qsort(edges, E, sizeof(struct Edge), compare);
27
28     // 2. Initialize parent array
29     for (int i = 0; i < V; i++)
30         parent[i] = i;
31
32     int minCost = 0, count = 0;
33
34     printf("Edges in MST:\n");
35     for (int i = 0; i < E; i++) {
36         int rootU = find(edges[i].u);
37         int rootV = find(edges[i].v);
38
39         // If including this edge doesn't form a cycle
40         if (rootU != rootV) {
41             printf("%d -- %d (Weight: %d)\n", edges[i].u, edges[i].v, edges[i].w);
42             minCost += edges[i].w;
43             parent[rootU] = rootV; // Union
44             count++;
45         }
46
47         if (count == V - 1) break; // MST complete
48     }
49
50     printf("Total Cost of MST: %d\n", minCost);
51 }
52
53 int main() {
54     int V = 5; // Vertices: 0, 1, 2, 3, 4
55     int E = 5; // Edges
56
57     // Graph edges (u, v, weight)
58     struct Edge edges[] = {
59         {0, 1, 10},
60         {0, 2, 6},
61         {0, 3, 5},
62         {1, 3, 15},
63         {2, 3, 4}
64     };
65
66     kruskal(edges, V, E);
67
68     return 0;
69 }
```

Output:

```
PS C:\Users\devan\OneDrive\Desktop> cd "c:\Users\devan\OneDrive\Desktop\"; if ($?) { gcc DEMO.c -o DEMO }; if ($?) { .\DEMO }
Edges in MST:
2 -- 3 (Weight: 4)
0 -- 3 (Weight: 5)
0 -- 1 (Weight: 10)
Total Cost of MST: 19
PS C:\Users\devan\OneDrive\Desktop>
```

Time Complexity Graph:

